Handout #27
Tuesday, 3 November 2026



Autumn Quarter 2026

Problem Set 6

Due: Tuesday, 17 November 2026 at 1:30pm via Gradescope
Topics: Interplanetary mission planning, porkchop plots

6.1) (10 points) Using a Lambert solver, one can estimate Voyager 2's heliocentric velocity vectors as it passed Jupiter (1979) , Saturn (1981), and Uranus (1986) on its way to Neptune. For each swingby, these can then be converted to planet-relative velocities. For the swingby of Jupiter (μ = 126,686,511.0 [km^3/sec^2]), assume the Jupiter-relative asymptotic arrival and departure velocity vectors were:

$$^{Jupiter}\underline{v}_{-\infty,heli} = \begin{bmatrix} -0.607781 \\ +7.832882 \\ -0.592059 \end{bmatrix} \left[\frac{\text{km}}{\text{sec}}\right] \text{ and } ^{Jupiter}\underline{v}_{+\infty,heli} = \begin{bmatrix} -7.610123 \\ -1.442188 \\ +0.636245 \end{bmatrix} \left[\frac{\text{km}}{\text{sec}}\right]$$

(These Jupiter-relative velocities are expressed in heliocentric coordinates and don't need to be transformed for this problem.) If this is modeled as an unpowered swingby with $^{Jupiter}v_\infty^2 = \left|^{Jupiter}\vec{v}_{-\infty}\right|\left|^{Jupiter}\vec{v}_{+\infty}\right|$, one can estimate the radius (not height) of closest approach to be r_p = 701,882.6 [km] and, of course, ΔV = 0 [km/sec].

In Lecture 9 we discussed powered gravity assists ("powered swingbys"). Because the magnitudes of the two velocity vectors above aren't exactly equal, this event would be more accurately modeled as a *powered* swingby. (Voyager 2's swingby was in fact unpowered, but its trajectory included maneuvers that we aren't modeling here.)

Reminder: the angle between two vectors can be related to the dot product of the two vectors.

Using the iterative method discussed in class and MATLAB's **`fzero`** function, find the radius of closest approach r_p and (scalar) ΔV for this powered swingby.
Use **`options = optimset('TolX', 1e-12);`** with **`fzero`** in order to solve for r_p to within 1 [m].

6.2) (50 points total) Here you will "reverse engineer" a number of quantities related to the launch asymptote and heliocentric Earth-to-Mars transfer trajectory for the Martian Moons eXploration (MMX) mission run by the Japan Aerospace Exploration Agency (JAXA). MMX departed from launch pad LA-Y2 at the Tanegashima Space Center in Japan aboard a Mitsubishi Heavy Industries H3 booster rocket on October 2026. It is now on its way to the Martian system, where it will attempt to sample regolith from Mars's largest moon, Phobos, and return it to Earth in 2031. For simplicity, you can ignore the details of arriving at and operating near Phobos and model this as an Earth to Mars mission. Base your estimates on the following data:

Launch from Tanegashima launch pad LA-Y2 was on
19 October 2026 at 19:41:03 UTC (20 October 2026 at 4:41:03 am Japan Standard Time)
and occurred at exactly the planned time (source: JAXA launch plan and news releases)

Arrival at Mars is planned for 15 August 2027 at 00:00:00 UTC (source: this is a "best guess," since JAXA's official materials don't specify an arrival date, and sources at collaborating organizations NASA and ESA state only "August 2027")

The Sun's gravitational parameter is $\mu_{Sun} = 1.3271244004193938 \times 10^{11}$ [km^3/sec^2]

1 [AU] = 149,597,870.7 [km]

Earth's obliquity to its orbit (the ecliptic) is 23.45 [deg]

Earth's rotation rate in inertial space is $\omega_{Earth} = 0.0000729211585530$ [rad/sec]

Greenwich sidereal time θ_g at the beginning of 2026 (1 January 2026 at 00:00:00 UTC)
was 6.7107222 [sidereal hours]

19 October 2026 was the 292nd day of 2026
(i.e. when October 19th began, 291 days had elapsed since the beginning of 2026)

Launch pad LA-Y2's *geocentric* latitude ϕ_L is +30.2333 [deg] (North latitude)

Launch pad LA-Y2's longitude λ_L is +130.9776 [deg] (East longitude)

a) (3 points) Use the **`AA279curtis_lambert.m`** solver for *prograde* trajectories to find the heliocentric asymptotic Earth departure velocity ${}^{Sun}\underline{v}_{+Earth,heli}$ and heliocentric asymptotic Mars arrival velocity ${}^{Sun}\underline{v}_{-Mars,heli}$. Both vectors are heliocentric asymptotic velocities expressed in heliocentric coordinates.

b) (7 points) Calculate the Earth-relative asymptotic departure velocity vector in geocentric (ECI) coordinates ${}^{Earth}\underline{v}_{+\infty,eci}$. (This will require finding Earth's heliocentric velocity and the probe's Earth-relative asymptotic departure velocity vector in heliocentric coordinates ${}^{Earth}\underline{v}_{+\infty,heli}$ as intermediate steps.)

c) (3 points) Estimate C_3 (in [km^2/sec^2]), RLA (in [deg]), and DLA (in [deg]) for this departure asymptote.

d) (3 points) Estimate MMX's launch azimuth and the orbital inclination into which it launched. (In your Lecture Notes on this topic, β = launch azimuth, ϕ_L = launch site *geocentric* latitude, α_∞ = RLA, and δ_∞ = DLA). You can assume the launch was in a prograde direction (i.e. 0 [deg] $\leq \beta \leq$ 180 [deg]). What would your estimate for launch azimuth be if the launch had been delayed by about 80 minutes to 21:00:00 UTC on 19 October 2026?

e) (3 points) For the actual 19:41:03 UTC launch time, does your estimated value of C_3 fall within a reasonable range for Earth-to-Mars missions? Is your estimated DLA less than or equal to the geocentric latitude of the launch site, indicating that JAXA avoided any "launch penalty" associated with launching to high-inclination orbits? How does your estimated launch azimuth compare with the planned easterly groundtrack shown on pg. 10 of the JAXA MMX launch plan? (See "JAXA MMX Launch Plan ANNOTATED BY BARROWS.pdf" on the AA 279B Canvas website in the "Problem Sets and Solutions" folder. This document is written in Japanese, but some relevant facts have been annotated in English!)

f) (3 points) Estimate MMX's Mars arrival excess speed $\left|{}^{Mars}\vec{v}_{-\infty}\right|$ by first converting the heliocentric asymptotic arrival velocity vector to a Mars-relative velocity.

g) (3 points) Use MATLAB's **`ode113`** propagator with **`options = odeset('RelTol', 1e-6, 'AbsTol', 1e-9);`** to simulate the Earth-to-Mars trajectory. This is a two-body trajectory on which only the Sun's gravity is active. Ensure your output datapoints from **`ode113`** are spaced 0.1 [day] apart or closer. Your result should include the following values stored at each datapoint in your simulation:

- Time in [JD] (Julian Date, including fractional part)
  This is technically called Spacecraft Event Time (SCET)
- The probe's heliocentric position vector in [km]
- The probe's heliocentric velocity vector in [km/sec]

h) (10 points) Step through your trajectory data and at each datapoint, compute and store the following additional data:

- Earth's heliocentric position vector in [km]
- Earth's heliocentric velocity vector in [km/sec]
- Mars's heliocentric position vector in [km]
- Mars's heliocentric velocity vector in [km/sec]
- Sun-Earth-Probe (SEP) angle in [deg]
  This is the angle between the Earth-to-Sun vector and the Earth-to-Probe vector. It is undefined when the Probe and Earth are at the same position. Probe-to-Earth communications are disrupted when SEP is small. (This is called a "solar conjunction." Because MMX is always farther away from the Sun than Earth, solar conjunctions also disrupt Earth-to-Probe communications. Additionally, when SEP reaches a maximum near 180 [deg], this is called a "solar opposition.")
- Sun-Probe-Earth (SPE) angle in [deg]
  This is the angle between the Probe-to-Sun vector and the Probe-to-Earth vector. It is undefined when the Probe and Earth are at the same position. Earth-to-Probe communications (the other direction of the link) are disrupted when SPE is small.
- Sun-Mars-Probe (SMP) angle in [deg]
  This is sometimes called the Sun phase angle and is the angle between the Mars-to-Sun vector and the Mars-to-Probe vector. It is undefined when the Probe and Mars are at the same position. The probe "sees" Mars as fully illuminated when SMP is small (assuming the Probe is between the Sun and Mars), and as unlit when SMP is near 180 [deg] (implying the probe is on the side of Mars opposite the Sun).

- Distance between the probe and Earth in [km]
- One-Way Light Time (OWLT) in [min]
  This is the time it takes light (at speed c = 299,792.458 [km/sec]) to cover the distance between the probe and Earth's center. (This is actually an approximation, since by the time a signal sent from the probe arrives at Earth, the Earth will have moved from where it was at the time the signal left the probe. This simplification is acceptable for our needs.)
- Earth-Received Time (ERT) in [JD] (Julian Date, including fractional part)
  This is the UTC time that a data signal sent by the probe at the corresponding SCET would be received at Earth. It is (SCET) + (OWLT in [days]).
- Spreading loss in [dB 1/m^2]
  This is $L_{spread} = \frac{1}{4\pi R^2}$ (undefined when R = 0), or more usefully for our purpose:

  $$L_{spread} = -70.99 - 20\log_{10} R_{km} \qquad \text{in } \left[\text{dB}\frac{1}{\text{m}^2}\right]$$

  where R is the Earth-probe distance in [km].

NOTE: There are many other items we are *not* having you compute due to the amount of extra work required. A typical mission analysis would include them, however, so here's a partial list for your amusement:

Maximum data rate given distance. (This requires more information on MMX's link budget than we could dig up. But, you should appreciate that maximum data rate decreases as Earth-probe distance and spreading loss increase.)

Transmission Time (TRM). This is the time that a data signal sent to the probe would need to be transmitted from Earth to arrive at the probe at the corresponding SCET. It is (SCET) – (OWLT in [days]).

Spacecraft visibility at various Deep Space Network (DSN) antenna sites. (This requires you to convert the probe's position from heliocentric to ECI to ECEF to South-East-Zenith to azimuth and elevation for a given DSN antenna to determine if a packet sent from the probe's position at time SCET can be "seen" at the antenna site at time ERT. This is critical for planning communications sessions with the probe.)

Received signal carrier frequency. (This includes Doppler shift of radio signals due to range rates and may impact receiver front end design and operation.)

i) (3 points) Plot the probe's trajectory on heliocentric x and y axes labeled in [AU]. On the same plot, show the Earth's locations on the launch and arrival dates along with Mars's locations on both dates. Does the end of the transfer match up with Mars's position when the probe arrives?

j) (3 points) Plot SEP angle versus time. Assuming that MMX-to-Earth communications are disrupted when SEP < 10 [deg], are there any times during the transfer when communications from MMX to Earth are disrupted? When do they begin and end (to the nearest day)?

k) (3 points) Plot SPE angle versus time on the same plot as part (j). Assuming that Earth-to-MMX communications are disrupted when SPE < 10 [deg], are there any times during the transfer when communications from Earth to MMX are disrupted? When do they begin and end (to the nearest day)?

l) (3 points) Plot SMP angle versus time. Are there any times during the transfer when Mars is fully illuminated as seen from the probe (to the nearest begin and end date)? Assume here that when SMP < 3 [deg], the full disk of Mars is illuminated as seen from the probe (assuming the probe is between the Sun and Mars) and that when SMP > 177 [deg], the probe is on the side of Mars opposite the Sun, and Mars is unlit as seen from the probe.

m) (3 points) When does the probe experience its farthest distance from Earth during the transfer (to the nearest day)? What are OWLT and spreading loss on this date?

6.3) (40 points total) For this problem you will create porkchop plots of important quantities for the 2026 Earth-to-Mars launch opportunity. The concept is to loop through a range of Earth departure and Mars arrival dates while recording data for later plotting. Many of the computations inside this loop are shared between this and Problem 6.2 parts (a) through (f).

Range of Departure and Arrival Dates

- Julian Dates for Earth departure should range from 2,461,261 [JD] through 2,461,421 [JD] (8 August 2026 12:00:00 UTC through 15 January 2027 12:00:00 UTC)
- Julian Dates for Mars arrival should range from 2,461,489 [JD] through 2,461,937 [JD] (24 March 2027 12:00:00 UTC through 14 June 2028 12:00:00 UTC)
- These date ranges include the actual MMX departure and arrival dates.
- For some launch opportunities, dates at the end of your Earth departure range could come *after* some of the dates at the beginning of your Mars arrival range. This happens to not be the case here. In general, for these non-sensical cases, you can record dummy values for C_3, RLA, DLA, and Mars arrival excess speed without having to do any Lambert or supporting calculations.

Grid Spacing

- Before even trying to fill out a two dimensional grid of candidate departure/arrival dates, try a single departure/arrival date combination with your code that finds C_3, RLA, DLA, and Mars arrival excess speed.
- The suggested Earth departure range is 160 days long and the Mars arrival range is 448 days long - both multiples of 32 days. This "powers of two" strategy lets you start with a coarse grid spacing of 32 days and then easily tighten it up later.
- Your porkchop plot contours should have some discernable structure even with 8-day spacing. For your final plots and analysis on parts (b) through (f), use a spacing of 2 days or shorter. Generating the grid data can take some time (my 6 year old laptop took 91 [sec] to generate the data for a 1-day spacing), so we suggest getting your code working well before generating finely spaced grids. You can then use MATLAB's **`save`** command to store this data so you don't have to constantly regenerate it while debugging your plotting code.

Additional Items

- Use the **`AA279curtis_lambert.m`** solver for *prograde* trajectories.
- If given a Julian Date, you can convert it to a textual format (year, month, day, hours, minutes, and seconds) using MATLAB's **`string`** and **`datetime`** functions.

a) (15 points) Build up the data matrices that will describe your porkchop plots. Loop through the specified ranges of Earth departure and Mars arrival dates and for each candidate pair of departure and arrival dates, store the loop variables:

- Julian Date of Earth departure
- Julian Date of Mars arrival

… and also compute and store:

- Offset Julian Date of Earth departure (days after nominal departure date)
  This is (Julian Date of Earth departure – 2,461,333) and is merely a suggestion that is easier to read on a plot than a 7-digit Julian Date
- Offset Julian Date of Mars arrival (days after nominal arrival date)
  This is (Julian Date of Mars arrival – 2,461,633) and is merely a suggestion that is easier to read on a plot than a 7-digit Julian Date
- Transfer time in [days]
  This is (Julian Date of Mars arrival - Julian Date of Earth departure)
- C_3 in [km^2/sec^2]
- RLA in [deg]
- DLA in [deg]
- Mars arrival excess speed $\left|{}^{Mars}\vec{v}_{-\infty}\right|$ in [km/sec]

b) (8 points) Plot contours of C_3 using MATLAB's **`contour`** or **`contourf`** function up through $C_3 = 40$ [km^2/sec^2]. On the horizontal axis should be Earth departure date (Offset Julian Date or Julian Date) and on the vertical axis should be Mars arrival date. Why does an area of higher C_3 separate the Type I trajectories from the Type II trajectories? Your Lecture 9 Notes include a porkchop plot of C_3 and Mars arrival excess speed for the 2018 Earth-to-Mars opportunity for reference. How does your porkchop plot compare with the one on pg. 12 of "NASA Earth-to-Mars Opportunities 2026-2045.pdf"? (This NASA Interplanetary Mission Design Handbook is available on the AA 279B Canvas website in the "Problem Sets and Solutions" folder.)

c) (3 points) Plot contours of RLA.
Compare your plot with the one on pg. 13 of the NASA Handbook.

d) (3 points) Plot contours of DLA.
Compare your plot with the one on pg. 13 of the NASA Handbook.

e) (3 points) Plot contours of Mars arrival excess speed up through 10 [km/sec].
Compare your plot with the one on pg. 14 of the NASA Handbook.

f) (8 points) Assume the MMX team specified a fixed Mars arrival date of approximately Julian Date 2,461,633.0 (15 August 2027 12:00:00 UTC). Also assume they determined that the earliest Earth departure date (beginning of the launch period) was approximately Julian Date 2,461,333.0 (19 October 2026 12:00:00 UTC).

Now, assume that the following three constraints must be satisfied by any usable combination of departure and arrival dates:

$$C_3 \leq 15.0 \left[{}^{km}/_{sec}\right]$$

$$\text{DLA} \leq 33.0 \left[deg\right]$$

$$\left|{}^{Mars}\vec{v}_{-\infty}\right| \leq 3.5 \left[{}^{km}/_{sec}\right]$$

What is the end date of the launch period? How many days long is the launch period? Which of the three constraints causes the launch period to close?